



Department of Computer Science and Engineering

Leading University

Project Report

on

EduVerse - A Smart Mobile Learning Platform

Course: Smartphone Application Development (SAD)

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ABSTRACT

EduVerse is a beginner-friendly mobile education application developed using Flutter and Dart. The main purpose of this project is to provide a simple digital learning environment where students can register, log in, view sample courses, manage their own questions, and check their profile information.

The application includes the core requirements of the Smartphone Application Development project: splash screen, authentication, profile page, CRUD operations, and an extra feature. Supabase is used as the backend service for authentication and database operations. The profile data and question data are stored in Supabase tables. Each logged-in user can only manage their own questions, which improves privacy and data separation.

The app uses simple Flutter widgets such as Scaffold, Card, ListView, TextField, ElevatedButton, FutureBuilder, and BottomNavigationBar. Reusable widgets are also used for common text fields, empty states, and profile information cards. Overall, EduVerse demonstrates basic mobile app development, backend integration, user authentication, database CRUD, navigation, and clean user interface design.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

EduVerse is a mobile-based education application prototype designed for students. The app focuses on simple learning support through course browsing, user authentication, profile management, and question management. It is developed as a Smartphone Application Development project using Flutter and Supabase.

The application starts with a splash screen and then checks the authentication state of the user. If the user is not logged in, the login page is shown. If the user is already logged in, the home page is opened. The home page contains three main sections: Courses, My Questions, and Profile.

The project is intentionally kept beginner-friendly. It does not use complex state management or advanced architecture. Instead, it uses clear screens, simple helper classes, reusable widgets, and direct Supabase database operations so that the project can be explained easily during viva.

1.2 Motivation

Education is becoming more digital and mobile-based. Students often need a simple platform where they can access learning content and manage their study questions. Many students also need an organized place to save their questions by subject so that they can revise them later.

The motivation behind EduVerse is to create a simple learning support app that combines authentication, profile information, course browsing, and question CRUD in one mobile application. This project also helps to practice real-world app development concepts such as backend connection, database table design, input forms, navigation, and secure user-based data access.

1.3 Objectives

- To develop a Flutter mobile application for education-related use.
- To implement login and registration using Supabase Authentication.
- To create a splash screen that introduces the app before opening the main flow.
- To display user profile information from the database.
- To implement CRUD operations for user questions.
- To add an extra feature through a static sample courses page.
- To use reusable widgets for repeated UI elements.
- To keep the interface simple, responsive, and easy to understand.

CHAPTER 2: BACKGROUND STUDY

2.1 Existing Applications

Several educational applications are already used by students, such as Google Classroom, 10 Minute School, Shikho, and other learning platforms. These apps support digital learning through video lessons, class materials, assignment sharing, and communication between students and teachers.

Google Classroom mainly focuses on classroom management, assignment sharing, and communication. 10 Minute School and Shikho focus more on learning content and course-based education. However, for this academic project, EduVerse is designed as a simplified student-centered prototype that demonstrates the required mobile application features within a small scope.

2.2 Project Planning

The EduVerse project was planned around the required SAD project features. The main modules are authentication, splash screen, profile page, CRUD operations, and an extra courses feature. Supabase was selected because it provides authentication and database services that can be connected easily with Flutter.

CHAPTER 3: PROBLEM STATEMENT AND OBJECTIVES

3.1 Problem Statement

Students often use separate tools for learning content, saving questions, and managing study information. This creates scattered data and makes revision difficult. A simple mobile app can help students keep their learning profile and study questions in one organized place. The problem addressed by EduVerse is the lack of a small, easy-to-use student learning support app that combines user authentication, course browsing, and personal question management.

3.2 Project Objectives

- Provide a secure registration and login system.
- Store each user profile with name, university name, and email.
- Allow each student to add, view, update, and delete only their own questions.
- Display sample courses as an extra feature of the app.
- Use bottom navigation for easy movement between main sections.
- Prepare the project in a way that is suitable for practical demonstration and viva.

CHAPTER 4: REQUIREMENT ANALYSIS

4.1 Functional Requirements

Requirement	Description
Splash Screen	The app shows an introductory EduVerse screen for about two seconds.
Authentication	Users can register and log in using email and password through Supabase.
Profile Page	The app displays full name, university name, email, and user ID.
CRUD Operations	Users can add, read, update, and delete their own questions.
Courses Page	Users can tap a course to view its details.
Logout	Users can securely sign out from the profile page.
Navigation	The app uses bottom navigation to switch between Courses, My Questions, and Profile.

4.2 Non-functional Requirements

- Usability: The interface should be simple and easy for students to use.
- Performance: The app should load pages smoothly and fetch data efficiently.
- Security: Each user should access only their own profile and questions.
- Maintainability: Code should be organized into pages, models, database helpers, and reusable widgets.
- Responsiveness: Screens should use scroll views, flexible layouts, and proper padding so the app can work on different device sizes.

4.3 System Requirements

- Operating System: Android device or emulator.
- Minimum RAM: 2 GB or higher recommended.
- Internet Connection: Required for Supabase authentication and database operations.
- Development Environment: Flutter SDK, Dart SDK, Android Studio or Visual Studio Code.

4.4 Software and Tools

Tool / Technology	Purpose
Flutter	Cross-platform mobile app development.
Dart	Programming language used by Flutter.
Supabase Authentication	Email and password login and registration.
Supabase Database	Storing profiles and questions.
supabase_flutter	Flutter package for connecting Supabase.
Git/GitHub	Version control and project backup.
VS Code	Code editing, running, and testing.

CHAPTER 5: METHODOLOGY

5.1 Development Methodology

The project was developed using a simple iterative method. First, the project requirements were identified. Then the Flutter project structure was created. After that, Supabase was connected for authentication and database support. The screens were implemented one by one: splash screen, login, register, home, courses, questions, and profile. Finally, CRUD operations and UI improvements were tested.

5.2 Application Architecture

EduVerse uses a simple folder-based structure. The auth folder contains authentication gate and services. The db folder contains database helper classes for profile and questions. The pages folder contains the main UI screens. The widgets folder contains reusable UI widgets. The models folder contains the CourseModel class.

- main.dart initializes Supabase and opens the SplashScreen.
- AuthGate checks whether a session exists and then opens LoginPage or HomePage.
- HomePage contains the BottomNavigationBar and switches between app sections.
- ProfileDb manages profile insert and read operations.
- QuestionsDb manages question select, insert, update, and delete operations.
- CustomTextField, EmptyStateWidget, and InfoCard are reusable widgets.
- CourseDetailsPage displays selected course information after a user taps a course card.

5.3 Database Design

The project uses two Supabase tables: profiles and questions. The profiles table stores user profile information. The questions table stores personal questions created by the logged-in user. The `user_id` field in questions links each question to the authenticated user.

Table: profiles

Column	Description
id	uuid primary key, same as Supabase Auth user id
full_name	text, not null
university_name	text, not null
email	text, not null
created_at	timestamp, default now

Table: questions

Column	Description
id	bigint identity primary key
user_id	uuid, current user id
title	text, not null
subject	text, not null
description	text, not null
created_at	timestamp, default now

CHAPTER 6: DIAGRAMS

6.1 Workflow Diagram

The workflow diagram shows how the app starts, checks authentication, and allows the user to access Courses, My Questions, and Profile after login.

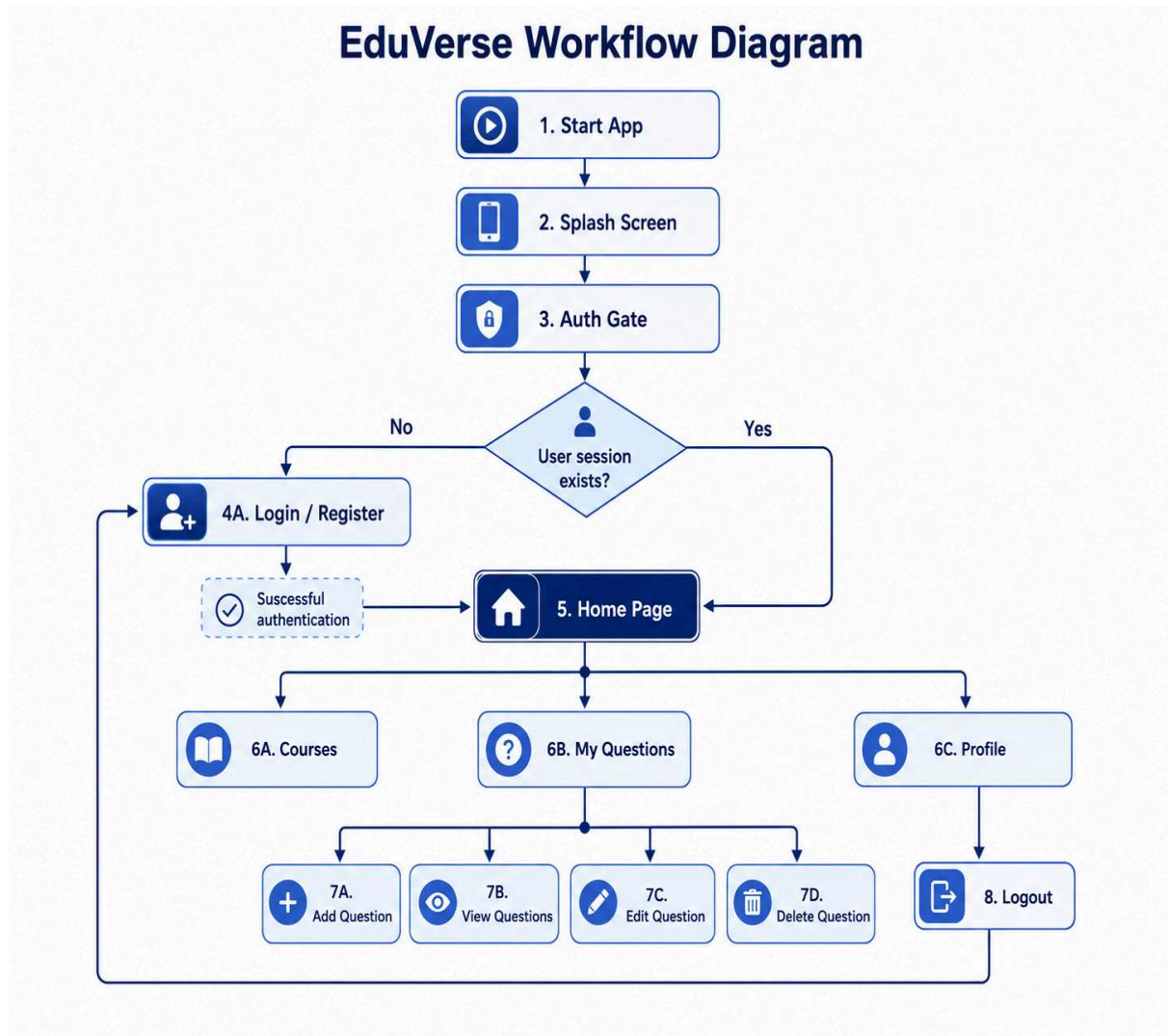


Figure 6.1: EduVerse Workflow Diagram

6.2 ER Diagram

The ER diagram shows the database relationship between the profiles and questions tables. The profiles table stores user information, while the questions table stores user-created questions. One profile can have many questions, and each question belongs to one profile through the user_id foreign key. This helps the system keep each user's questions separate and secure.

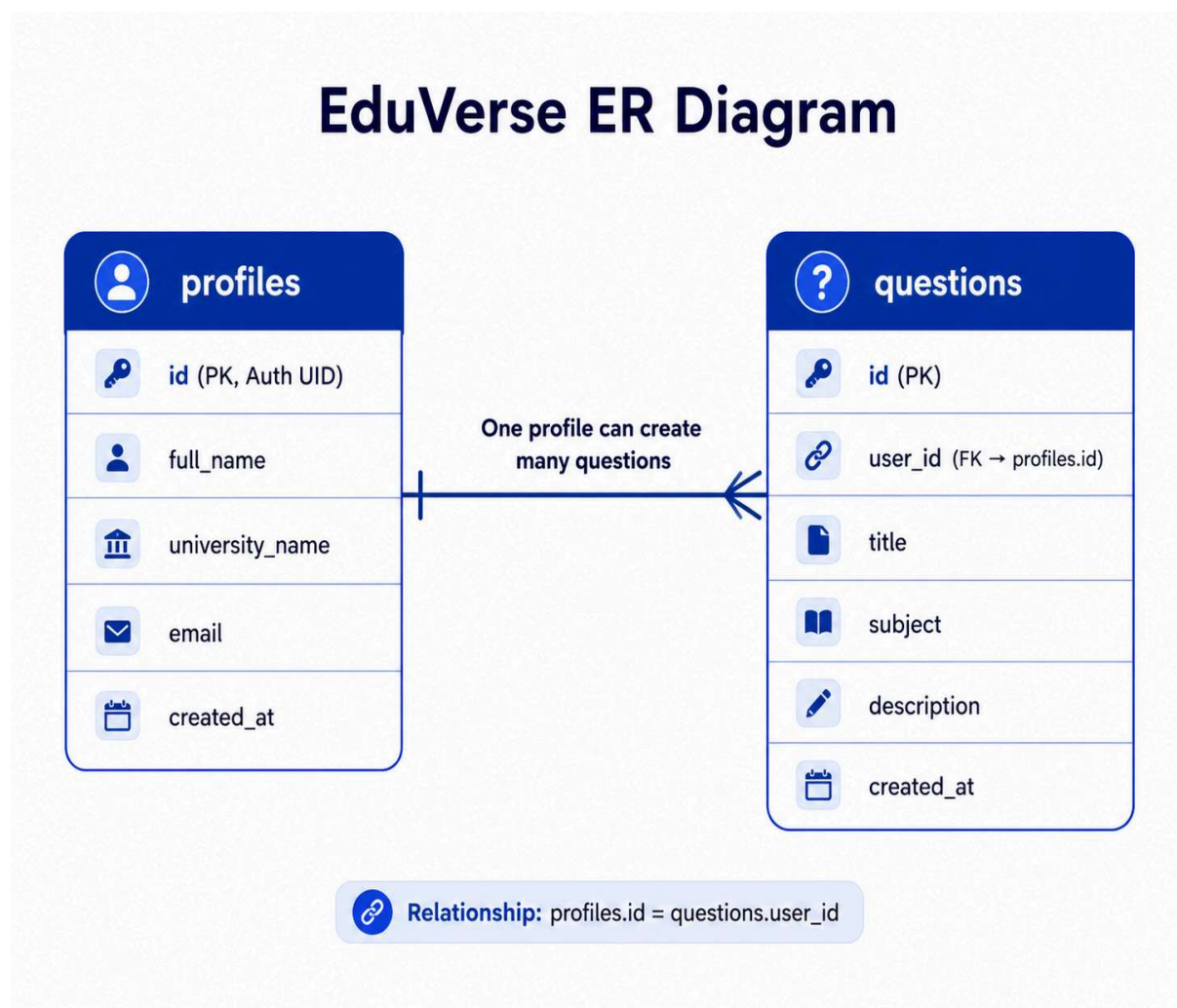


Figure 6.2: EduVerse ER Diagram

CHAPTER 7: SNAPSHOT OF COMPLETED FEATURES

This chapter presents visual snapshots of the completed EduVerse features. These screens are prepared from the implemented UI structure of the project.

7.1 Splash Screen

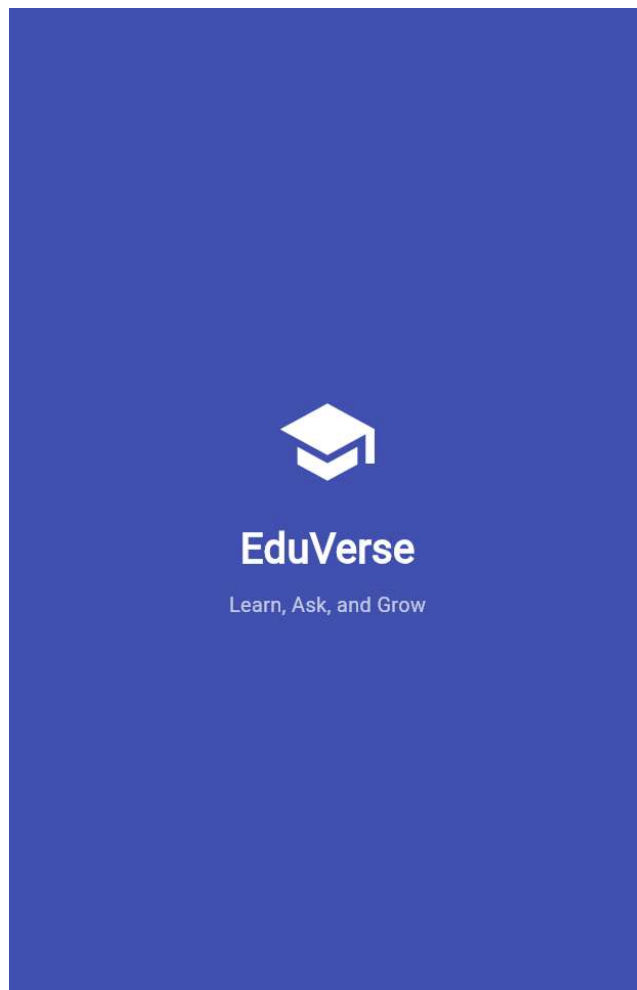
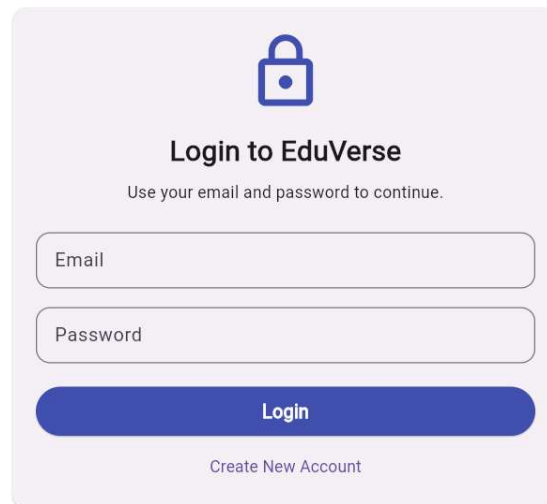


Figure 7.1: Splash Screen

The splash screen displays the app name EduVerse and tagline Learn, Ask, and Grow before going to authentication checking.

7.2 Login Screen

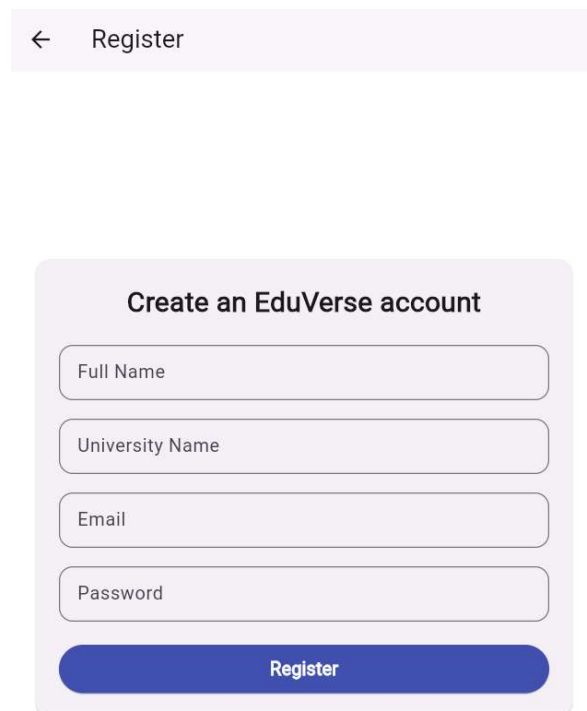


The login screen for EduVerse is a light purple rounded rectangle. At the top center is a blue padlock icon. Below it, the text "Login to EduVerse" is displayed in bold, followed by the instruction "Use your email and password to continue." in a smaller font. There are two input fields: "Email" and "Password", both with rounded corners and light purple borders. Below these fields is a prominent blue "Login" button with white text. At the bottom, centered, is a link that says "Create New Account" in a small, light purple font.

Figure 7.2: Login Screen

The login screen allows registered users to enter email and password and sign in through Supabase.

7.3 Register Screen



The image shows a mobile application interface for a registration screen. At the top, there is a light purple header bar with a left-pointing arrow and the text "Register". Below this is a white rounded rectangle containing the title "Create an EduVerse account". Under the title are four stacked input fields with labels "Full Name", "University Name", "Email", and "Password". At the bottom of this white box is a blue rounded button with the text "Register".

Figure 7.3: Register Screen

The register screen collects full name, university name, email, and password, then creates both auth account and profile row.

7.4 Courses Page

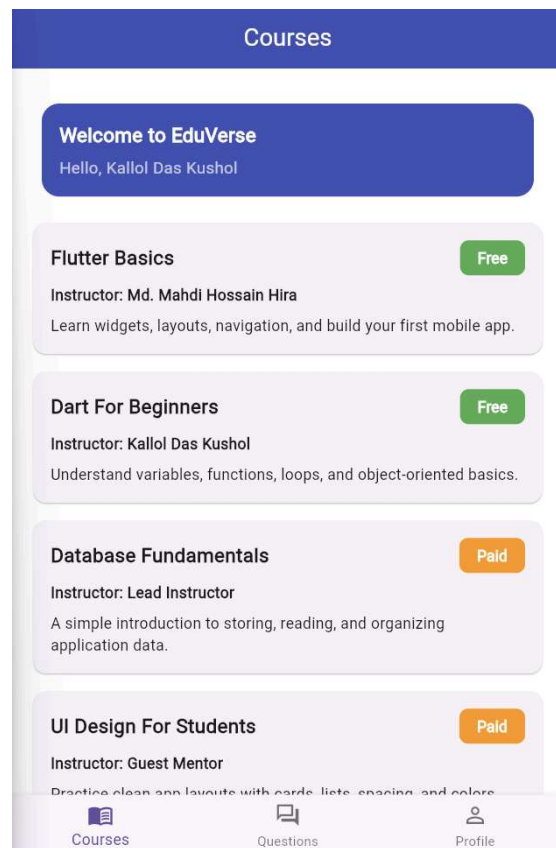


Figure 7.4: Courses Page

The courses page is the extra feature of the app. It displays static course cards with free and paid labels, and users can tap a course card to view detailed course information.

7.5 My Questions Page

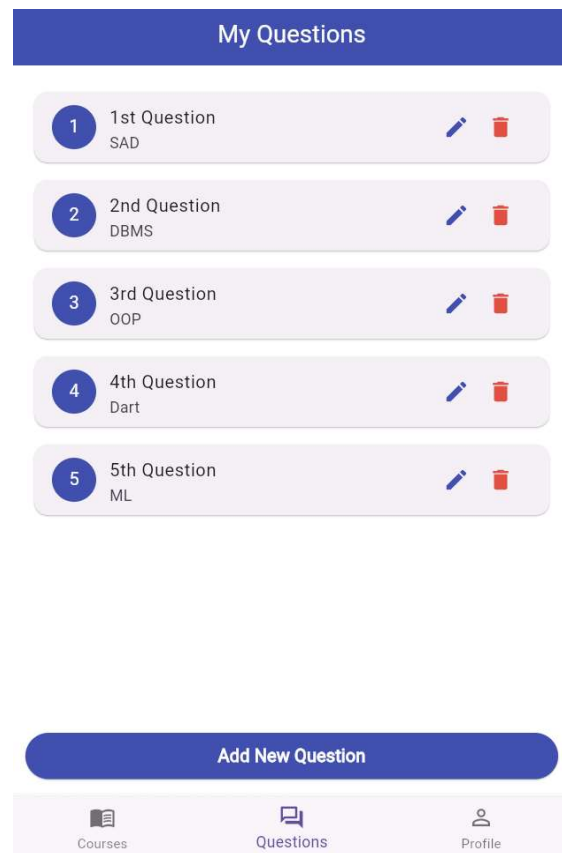


Figure 7.5: My Questions Page

The My Questions page implements CRUD operations. A user can add, read, update, and delete questions.

7.6 Profile Page

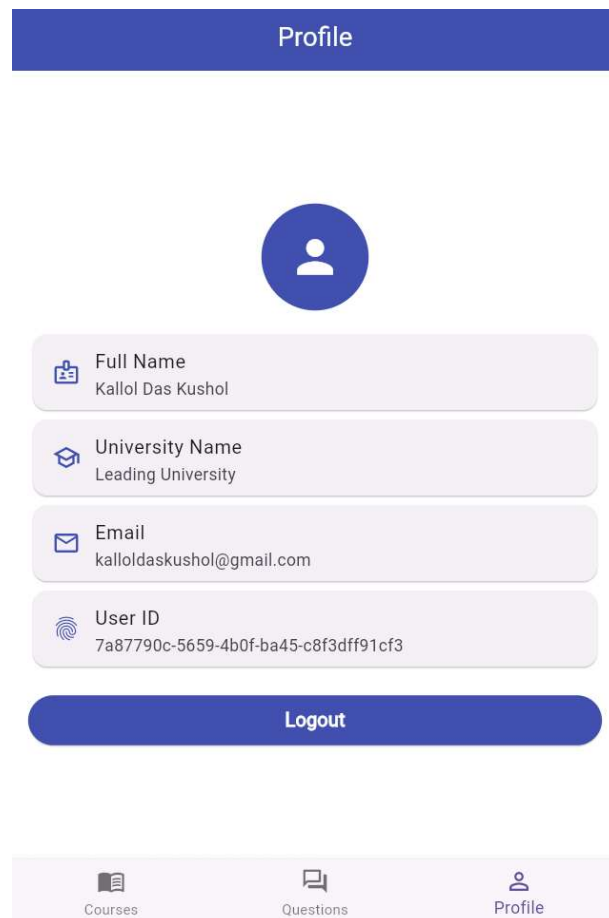


Figure 7.6: Profile Page

The profile page displays user information from the profile table and includes a logout button.

CHAPTER 8: LIMITATIONS AND FUTURE WORK

8.1 Limitations

- The courses are static and are not stored in the database.
- The app does not include video lessons in the current implementation.
- The profile page is read-only and does not support profile editing.
- There is no teacher dashboard or admin panel.
- There is no payment gateway for paid courses.
- The app needs more testing on different devices and screen sizes.

8.2 Future Work

- Store courses and lessons in Supabase database.
- Add video lesson page and lesson completion tracking.
- Add assignment submission feature.
- Add discussion or chat feature for student interaction.
- Add notification system for deadlines and course updates.
- Add profile editing and image upload.
- Add teacher/admin panel for managing courses and questions.
- Improve UI responsiveness and add more validation.

CHAPTER 9: CONCLUSION

EduVerse is a practical mobile education application prototype built using Flutter and Supabase. It successfully implements the main SAD project requirements: splash screen, authentication, profile page, CRUD operations, and an extra feature.

The project demonstrates important mobile app development concepts such as screen navigation, form input, reusable widgets, backend connection, user authentication, database table design, and user-specific data access. The My Questions feature is the main CRUD module, and the Courses page works as an extra educational feature.

Although the current version is simple, it can be expanded into a larger learning platform in the future. Features such as dynamic courses, video lessons, assignments, progress tracking, discussion, and notifications can make EduVerse more complete and useful for students.

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